

Enclosure Specification

CJ2 Voice Emergency Checklist Unit

Cessna Citation CJ2 (CE 525A) — bench / training prototype

Two-piece housing: DZUS-rail panel bezel + remote processor box

Mounting, materials, audio, thermal, environmental targets,
labeling, deliverables, and reference dimensions.

DEMO / TRAINING USE ONLY — NOT FOR ACTUAL FLIGHT OPERATIONS

Not a certified avionics installation. Borrows airworthiness conventions (DZUS, DO-160) so the prototype fits and looks right, but it is a mock-up. Real installation requires a licensed A&P/IA and FAA approval (337/STC/TSO).

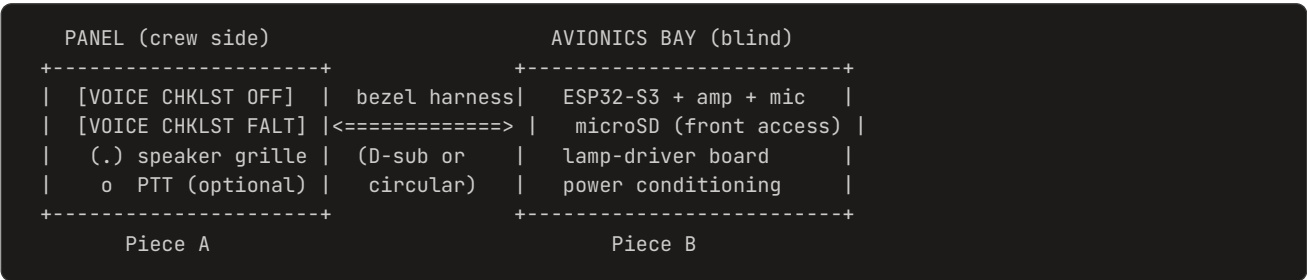
1 Design Intent & Architecture

This document is written for the mechanical / industrial-design engineer who will model, prototype, and fabricate the housing for the voice emergency-checklist unit. It assumes the electronics, pin-out, and behavior are fixed and documented in the companion `TECH_DATA.md` and the wiring diagram (reproduced in §9).

The unit splits into two physical pieces joined by a short harness, because the pilot-facing controls and the processor have very different placement needs:

Piece	Contents	Where it lives	Why
A. Panel bezel (pilot-facing)	Split-legend annunciator switch, speaker + grille, optional PTT button	Front instrument panel / pedestal, on the DZUS rail	Must be seen and reached by the crew; the dark-cockpit annunciator must sit in the pilot's normal scan
B. Remote processor box (blind)	ESP32-S3 board, INMP441 mic, MAX98357A amp, microSD, lamp-driver board, power conditioning	Avionics bay / behind-panel shelf, blind-mounted	Keeps heat, the SD slot, and wiring out of the panel; only the bezel needs panel real estate

A single all-in-one panel box is acceptable for a pure bench demo, but the two-piece split is the recommended target: it mirrors how real remote-mount avionics are installed and keeps the microphone away from cooling-fan and avionics noise.



2 Piece A — Panel Bezel

2.1 Form factor & mounting

- Mounting standard: DZUS rail (the CJ2 pedestal and most of its panel use DZUS). Fastener pitch (hole-to-hole) = 3/8 in (9.525 mm); fastener clearance hole 0.255 in (6.48 mm).
- Bezel height must be a whole multiple of 3/8 in (the “DZUS rhythm”). Target a 3-unit panel = 1.125 in (28.575 mm) tall, or 4-unit = 1.5 in (38.1 mm) if the speaker grille needs more room.
- Standard pedestal panel width = 5.75 in (146.05 mm) nominal aluminum ($\approx$ 144.45 mm usable face). Use this width so the bezel drops into a standard slot.
- Backplate: 1/16 in (1.6 mm) aluminum (6061-T6) per DZUS convention; first / last fastener centers $1.5 \times 3/8$ in = 14.29 mm from each end.
- Alternative (round-hole) mount: if no DZUS slot is available, provide a variant that fits a standard 3-1/8 in (79.4 mm) instrument cutout with four 6-32 screws on the standard bolt circle, so it can replace a blanking plate.

2.2 Front-face layout (top → bottom, dark-cockpit order)

- Split-legend annunciator switch (Applied Avionics VIVISUN / Korry). Cutout per the specific switch series' datasheet — confirm the exact part before cutting. A typical VIVISUN 1-pole rectangular bezel is $\approx 15 \times 15$ mm to 19×19 mm face; provide snap or screw retention per that series.
- Legend engraving: top half = VOICE CHKLST OFF (white); bottom half = VOICE CHKLST FAULT (amber). Legend orientation must read upright when panel-installed.
- Speaker grille (see §4) — open area $\geq 40\%$ over the speaker cone; offset from the switch so a finger on the switch never covers the grille.
- PTT button (optional) — momentary, guarded or recessed so it is not bumped. If the aircraft already has a yoke / PTT tie-in, omit this and route PTT through the harness.

2.3 Bezel material & finish

- Face: 6061-T6 aluminum, 2.0–3.0 mm, or ABS/PC for a non-structural demo.
- Finish: matte black, low-gloss (≤ 10 gloss units) to suppress windshield glare — standard for instrument-panel parts and consistent with dark-cockpit philosophy.
- Engraving: laser-etched + paint-fill in white and amber to match the legend colors, OR rely solely on the switch's internal legend (preferred — fewer painted parts to chip).
- Edges chamfered 0.5 mm; no sharp corners facing the crew.

3 Piece B — Remote Processor Box

3.1 Size & internal layout

- Size the internal volume around the largest board path = the ESP32-S3 DevKitC-1 ($\approx 70 \times 26$ mm) plus the MAX98357A breakout, mic, microSD breakout, and the 2-channel lamp-driver board. Practical outer envelope: $\approx 110 \times 80 \times 45$ mm (L $\times$ W $\times$ H). Confirm against the actual stacked board set.
- Use internal standoffs / PCB rails (M2.5 brass inserts) so boards are screwed down, not floating — important for the vibration environment (§6).
- Keep the INMP441 mic away from the amplifier and any fan. If the mic lives in this box, add an acoustic port (small grille or 4–6 mm hole with mesh) on a quiet face. For best recognition the mic can instead live in the bezel near the crew — engineer's choice — but keep the I2S run short (< 150 mm).

3.2 Access & connectors

- microSD access: a slot on a removable face or an externally-accessible push-push SD carrier, so the card can be swapped (to reconfigure the aircraft) without opening the sealed box. Label it CONFIG CARD – FAT32.
- USB-C service port: a covered / recessed USB-C pass-through for flashing and power on the bench. Add a silicone plug or hinged cover; it is not for in-service use.
- Main connector: one circular bayonet connector (e.g. a small MIL-style / M12) OR a 9-pin D-sub carrying: bezel switch SELECT, the two legend-lamp drives, PTT, speaker +/-, and power / ground. Pin-out to be finalized from board_pins.h. Use a keyed, positive-latching connector — no loose flying leads.
- Power: accept bench supply (USB 5 V ≥ 1 A). If a 28 VDC aircraft-bus mock-up is wanted, include a 28 V $\rightarrow$ 5 V DC-DC (≥ 2 A) inside the box with input transient protection (TVS + fuse) — but mark clearly that this is a demo regulator, not DO-160 qualified.

3.3 Box material & EMI

- Material: aluminum (extruded or machined) preferred for the remote box — it doubles as an EMI shield and a heatsink. ABS/PC acceptable for a pure desk demo.
- If plastic, add a conductive shield liner or coating (nickel paint) tied to ground, since the unit sits near sensitive avionics; good practice even for a mock-up.
- Single-point chassis ground stud bonded to the connector shell and the ESP32 ground.

4 Audio (Speaker) Details

- Speaker: 4–8 Ω , ≥ 2 W, sealed-back or with a small rear volume (5–15 cm³) to avoid a tinny sound; the MAX98357A peaks ~ 650 mA at 5 V / 4 Ω .
- Grille: perforated metal or molded slots, $\geq 40\%$ open area, with a thin acoustic mesh behind it for dust. Keep the speaker front-firing toward the crew.
- Gasket the speaker to the bezel to prevent buzz at volume.

5 Thermal

- The ESP32-S3 running ESP-SR is the main heat source but is low power (a few hundred mW typical); no fan is required.
- Provide passive convection: a few vent slots low and high on the remote box, or bond the ESP module's ground plane to the aluminum wall with a thermal pad.
- Do not seal the box fully airtight if relying on convection. If sealing is required for dust / water, switch to conduction cooling (thermal pad to the aluminum case) and verify the internal rise stays < 20 °C above ambient at 55 °C ambient.
- Keep the DC-DC (if fitted) on its own thermal path; it is the second heat source.

6 Environmental Targets

DO-160G categories a real unit in this location would target. For the prototype, treat these as design guidance, not test obligations.

DO-160G section	Category to design toward	Note for this unit
§4 Temperature / Altitude	Cat A2 (controlled cockpit)	-15 °C to +55 °C operating; pressurized cabin
§7 Shock & crash safety	Operational shock + crash-safety	Mounts / standoffs must retain boards; nothing becomes a projectile
§8 Vibration	Cat S (fixed-wing)	Locking hardware; conformal-coat the boards; no press-fit-only parts
§6 Humidity	Cat A	48-hr; conformal coat recommended
§16/17 Power input / spike	—	Add TVS + fuse on any 28 V input mock-up
§20/21 RF emission / susceptibility	—	Aluminum case + grounded shield; Wi-Fi stays disabled (it is, in firmware)
§25 ESD	—	Bond exposed metal; recessed connectors
§26 Flammability	—	Use flame-rated plastics (UL94 V-0) if not aluminum

The firmware already runs fully offline with Wi-Fi and Bluetooth off, which materially helps the RF-emission story for any future qualification.

7 Labeling & Markings

- Required placard on the bezel and the box: DEMO / TRAINING ONLY — NOT FOR FLIGHT.
- Box exterior: unit name, a serial / asset field, the config-card format (FAT32), and the USB service-port “bench use only” note.
- Annunciator legends: VOICE CHKLST OFF (white, top), VOICE CHKLST FAULT (amber, bottom).
- Use amber for caution / white for status, consistent with FAA AC 25-11B color conventions used elsewhere in this project.

8 Deliverables Requested from the Engineer

- 1 3D CAD (STEP + native) of both pieces, with the board set and switch modeled in place.
- 2 2D drawings with the DZUS hole pattern, the switch cutout (dimensioned from the chosen switch datasheet), grille pattern, and connector cutouts — fully dimensioned, GD&T where it matters (switch cutout, DZUS holes).
- 3 Connector pin-out mapping the bezel-to-box harness to board_pins.h signals.
- 4 A bench-printable prototype (FDM/SLA) of both pieces for fit-check before any aluminum.
- 5 Mechanical BOM (fasteners, inserts, connector, speaker, grille mesh, gasket).
- 6 Tolerance call-outs: switch cutout ± 0.1 mm; DZUS holes ± 0.1 mm on the 9.525 mm pitch; general ± 0.25 mm.

8.1 Open items for the engineer to confirm

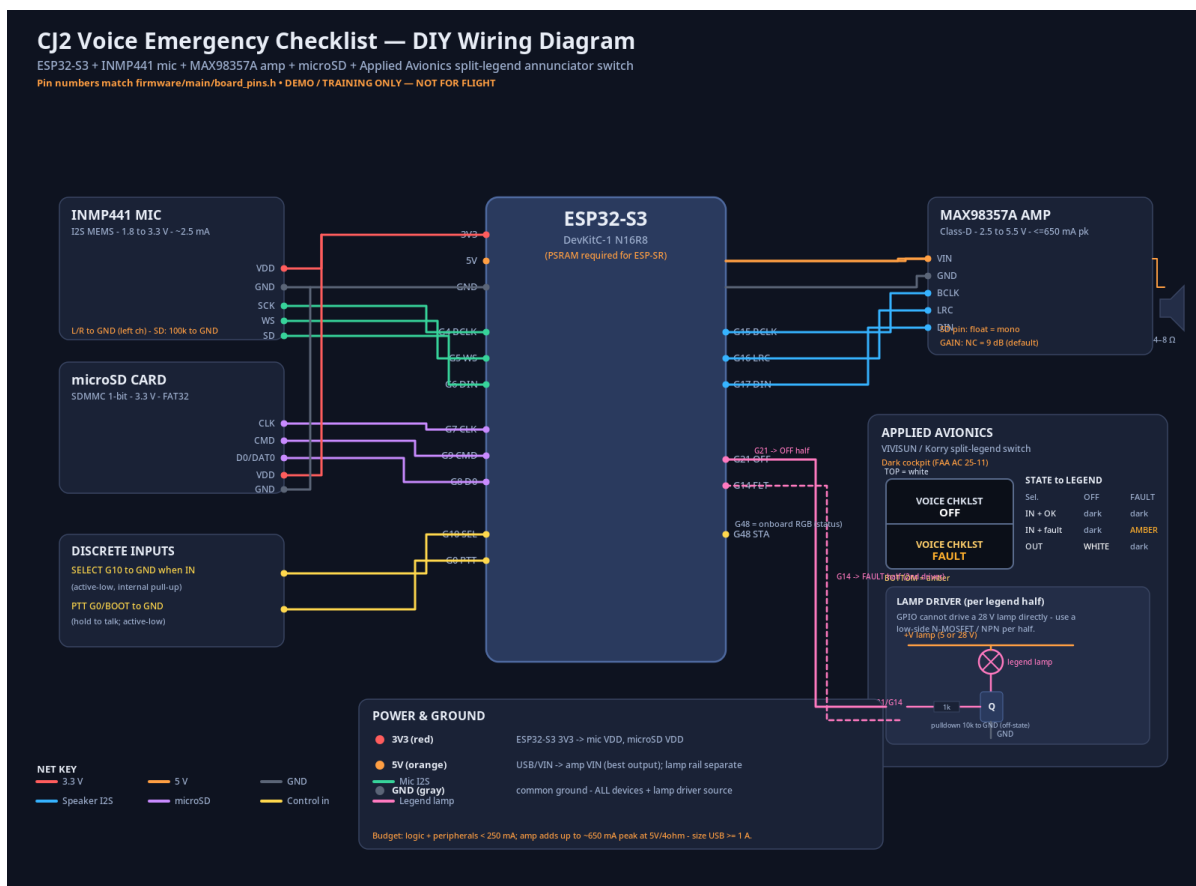
The single most critical unknown is the exact Applied Avionics switch part number — it drives the bezel cutout and depth. Nothing else can be finalized until it is fixed.

- Mounting reality in the target panel: is there a free DZUS slot, or must this go into a 3-1/8 in round hole or a custom sub-panel? Pick the §2.1 variant accordingly.
- Where the mic lives (bezel vs. remote box) — affects acoustic ports and harness count.
- Whether a 28 V input is wanted, or USB 5 V bench power is sufficient for the demo.
- Speaker model — sets the grille open area and the rear-volume cavity.

9 Reference Dimensions & Wiring Context

Item	Value	Source
DZUS fastener pitch	3/8 in = 9.525 mm	DZUS rail standard
DZUS clearance hole	0.255 in = 6.48 mm	DZUS rail standard
Panel backplate thickness	1/16 in = 1.6 mm	DZUS convention
First fastener offset from edge	1.5 × pitch = 14.29 mm	DZUS convention
Standard pedestal panel width	≈ 146 mm alu (144.45 mm face)	DZUS panel convention
Standard round instrument hole	3-1/8 in = 79.4 mm	Instrument panel standard
Remote box target envelope	≈ 110 × 80 × 45 mm	This design (confirm vs. boards)
Speaker	4–8 Ω, ≥ 2 W	TECH_DATA.md §3.2

The harness between Piece A and Piece B carries the signals shown below. The full electrical reference (pin map, per-device wiring, lamp driver) is in the companion technical packet.



System wiring diagram (from the companion technical packet).

10 Source References

#	Reference	Source
1	DZUS rail dimensions & the “DZUS rhythm”	MyCockpit panel-building guide
2	DZUS rail product reference	Dallas Avionics
3	Standard 3-1/8 in instrument cutout / adapter	Aircraft Spruce
4	DO-160G environmental categories (summary)	Wikipedia
5	Environmental qualification guidance	FAA AC 21-16G
6	Annunciator color / dark-cockpit conventions	FAA AC 25-11B
7	LED lighted pushbutton / split-legend switches	Applied Avionics (VIVISUN)

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